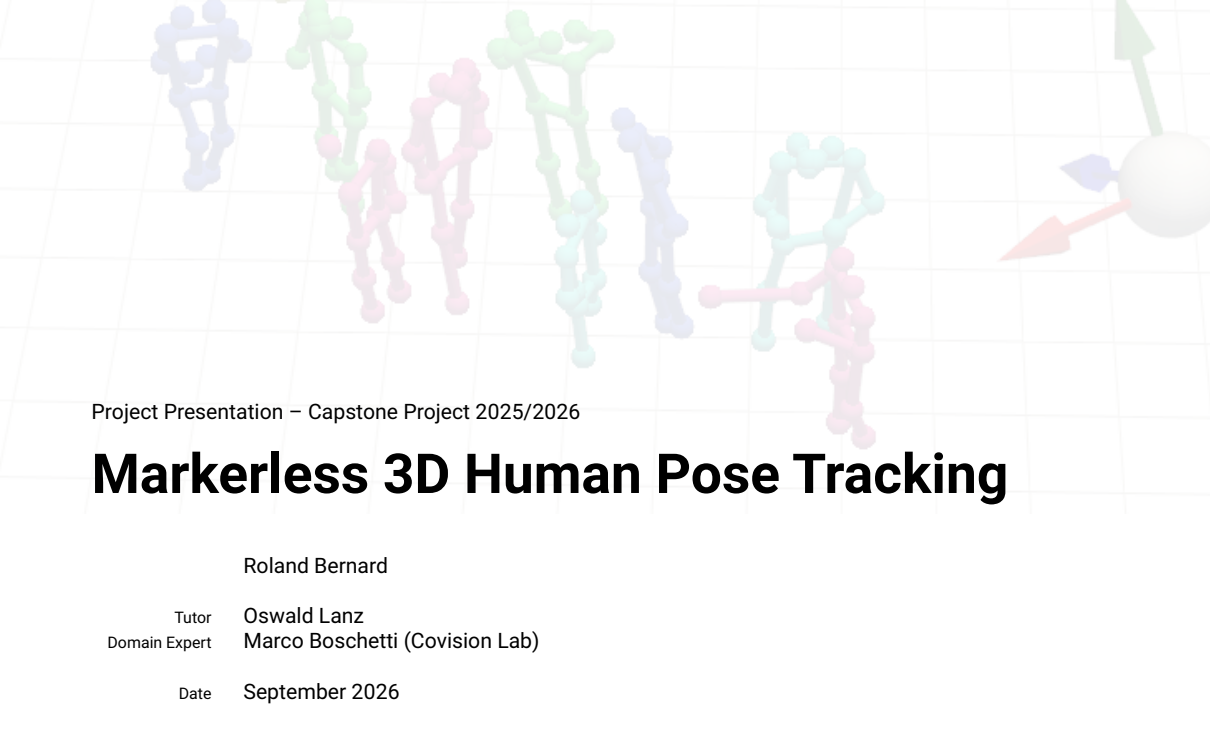




Project Presentation – Capstone Project 2025/2026

Markerless 3D Human Pose Tracking

Roland Bernard

Tutor Oswald Lanz
Domain Expert Marco Boschetti (Covision Lab)

Date September 2026

Introduction

- ▶ Problem: Track poses of multiple people in videos across multiple views.
- ▶ Possible Applications:
 - Robotics and human-robot interactions.
 - In healthcare, for tasks such as gait analysis or remote patient monitoring.
 - For real-time motion capture or natural body-based control interfaces for gaming.
- ▶ Objectives of this project:
 - Develop and evaluate a 3D pose tracking pipeline.



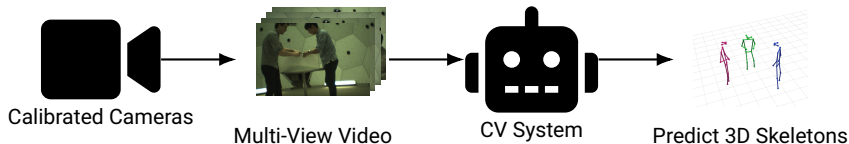
Task Formulation

- ▶ Given:

- Multiple video sequences containing multiple people.
- Recorded synchronously from multiple camera views.
- Camera calibration parameters, including extrinsics, intrinsic and distortion.

- ▶ Output:

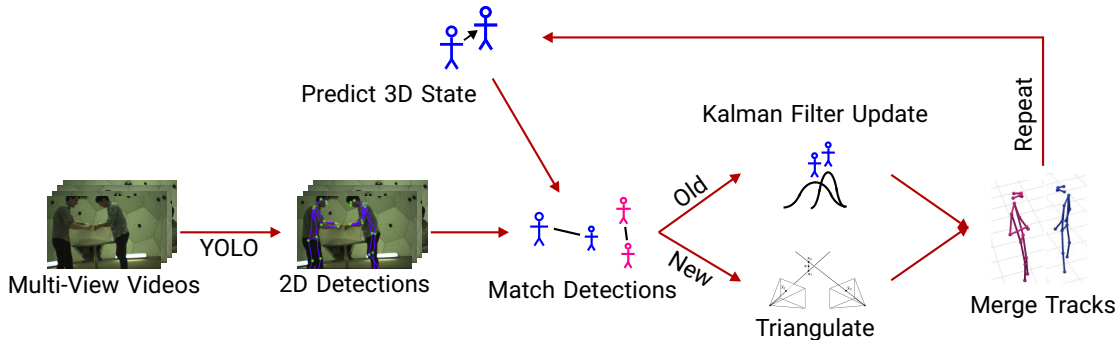
- 3D coordinates of skeletal joints for each subject.
- Temporally consistent movement of skeletal joints over time.



System Architecture – Overview

► Summary:

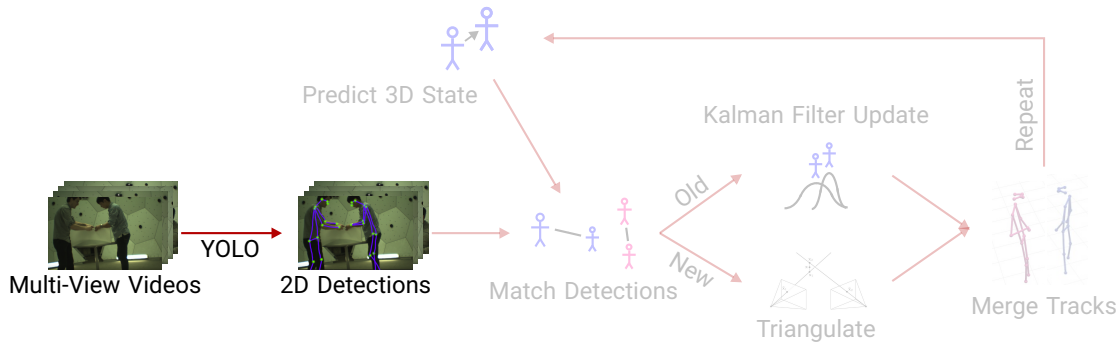
- Perform single-view single-frame 2D estimation using pretrained model.
- Cross-view and prediction-view matching of detections using Mahalanobis distance.
- Fusion of 2D observations into 3D skeletal trajectories using extended Kalman filter.



System Architecture – 2D Detections

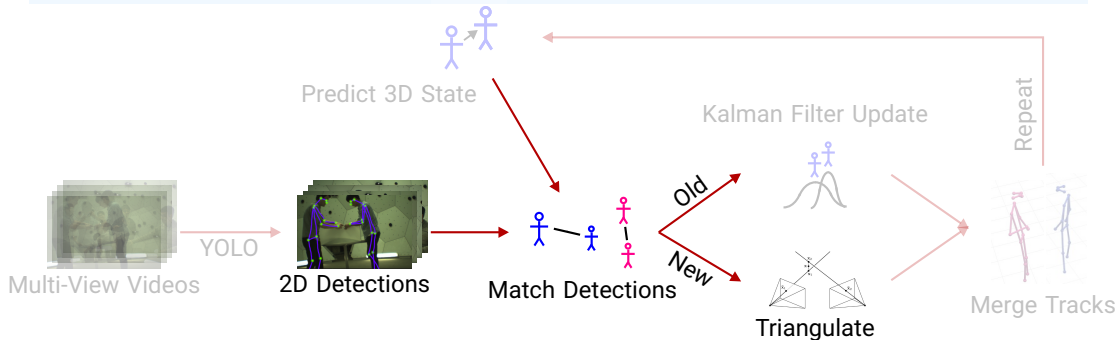
► Summary:

- Using YOLO26-pose family of models for the detector.
- Dynamically scale observation covariances based on YOLO keypoint confidence.
- Transform 2D pixel coordinates into normalized camera coordinates.



System Architecture – Data Association

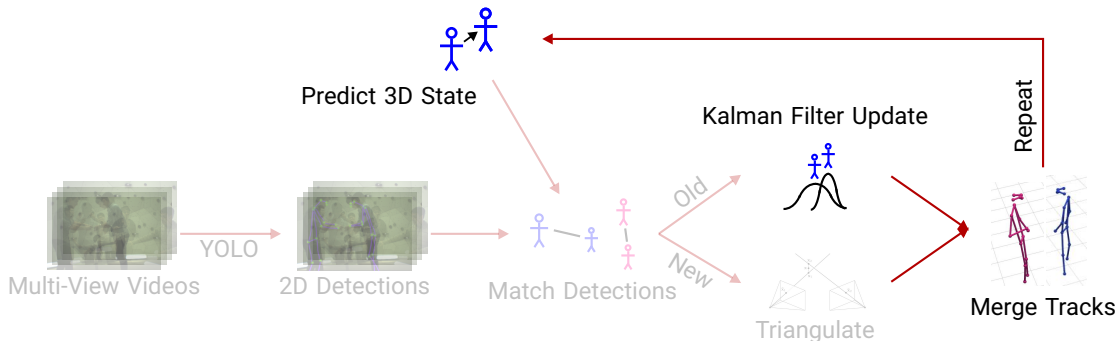
- ▶ Two-step process:
 - For each view, try to associate detections to predicted 3D state.
 - For all unmatched detections, try to associate between different views.
- ▶ For all cross-view matches:
 - Triangulate 3D keypoint locations using all matched views to initialize tracks.



System Architecture – Extended Kalman Filtering

► Summary:

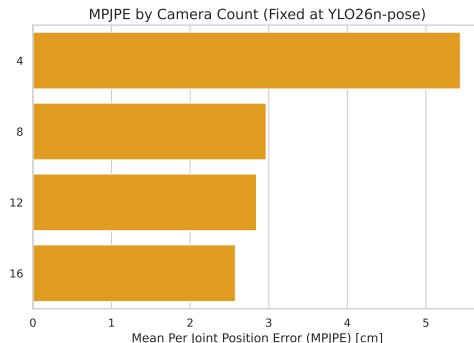
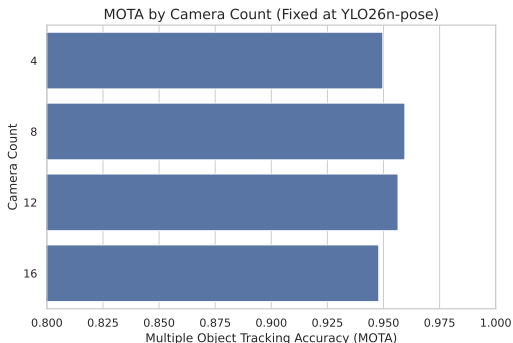
- State representation contains 3D position and velocity for each keypoint.
- Prediction uses standard linear velocity model.
- All associated observations are concatenated and considered at once.
- Since the observation process is non-linear extended Kalman filtering is used.



Skeletal Constraints

► Summary:

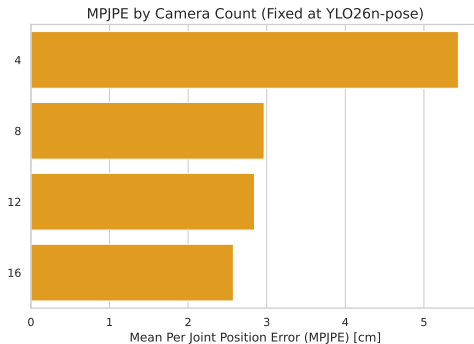
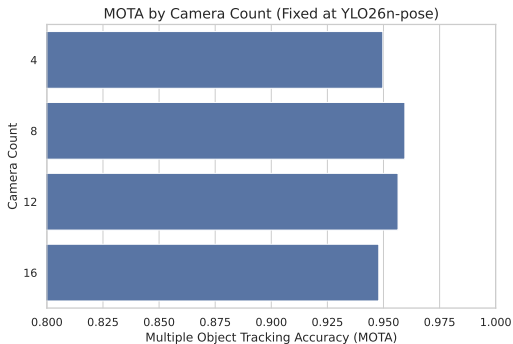
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Learning Parameters from Data

► Summary:

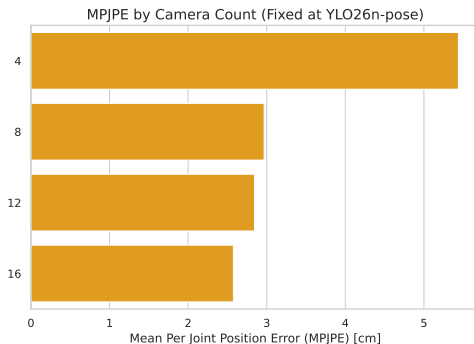
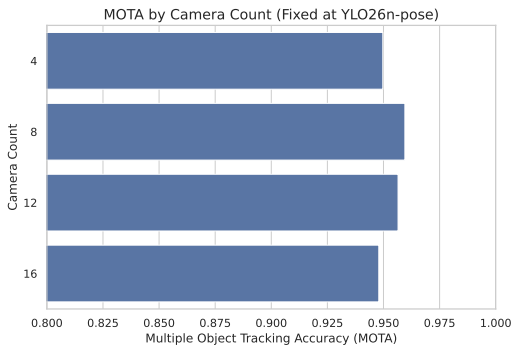
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Learning Parameters from Data – Custom YOLO Head

► Summary:

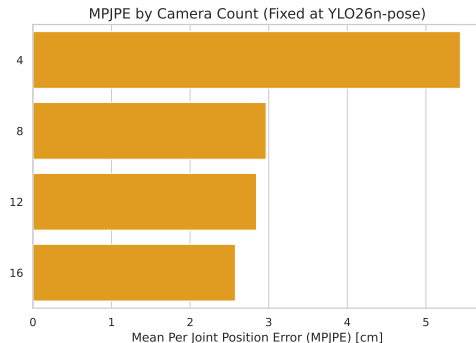
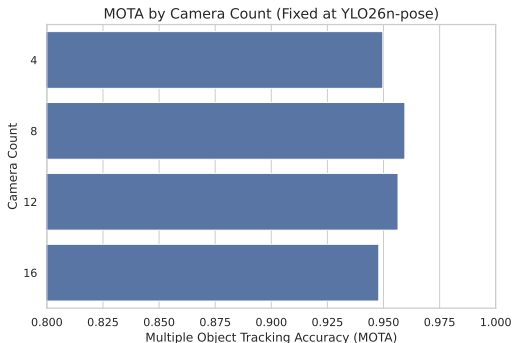
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- Consequuntur repudiandae non odio sunt consequatur alias.
- Magni esse et non quas eveniet molestias quidem aut.
- Id ut nihil aut et numquam.



Learning Parameters from Data – Kalman Filter

► Summary:

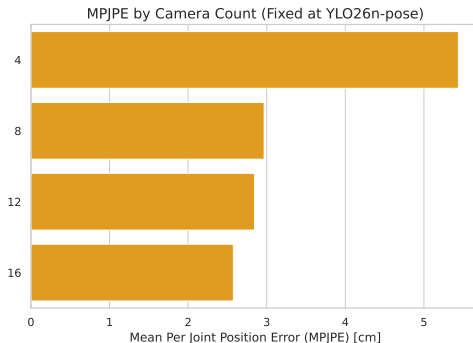
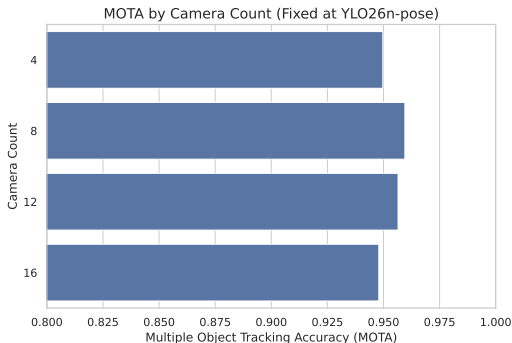
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- Consequuntur repudiandae non odio sunt consequatur alias.
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- Id ut nihil aut et numquam.



Learning Parameters from Data – Other Parameters

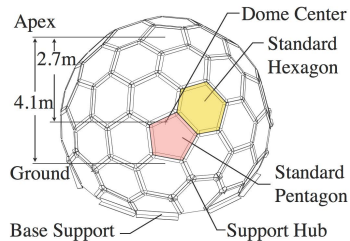
► Summary:

- Ipsum laborum incidunt consequatur nobis.
- Consequuntur repudiandae non odio sunt consequatur alias.
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Dataset Overview – CMU Panoptic Dataset

- ▶ Source:
 - Video captured in a specialized 5.49m diameter dome-shaped room.
 - Equipped with hundreds of inward facing cameras attached to the outside panels.
- ▶ Format and Data Characteristics:
 - 84 video sequences totaling approximately 11.5 hours of video.
 - VGA cameras recording at a 640×480 resolution at 25 fps.
 - Includes ground truth 3D skeletons and tracks in COCO19 format.



Experimental Setup

- ▶ **Setup:**

- Run evaluation on a subset of 8 sequences from the dataset.
- Compare generated pose tracks against ground truth annotations.
- Evaluate using varying number of camera views and differently sized YOLO models.
- Also record timing statistics for evaluation of algorithm efficiency.

- ▶ **Evaluation Metrics:**

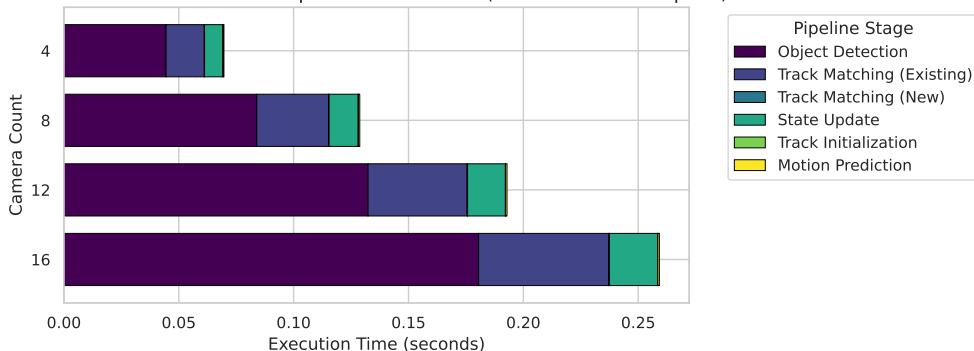
- Mean Per-Joint Position Error (MPJPE).
- Multi-Object Tracking Accuracy (MOTA).
- Also collected Precision and Recall.

Quantitative Results – Runtime Efficiency

► Main Finding:

- Detection is the largest part of the execution time per frame.
- Execution time of detection scales linearly with number of cameras.
- Matching and filtering also scale with camera count but slight less quickly.

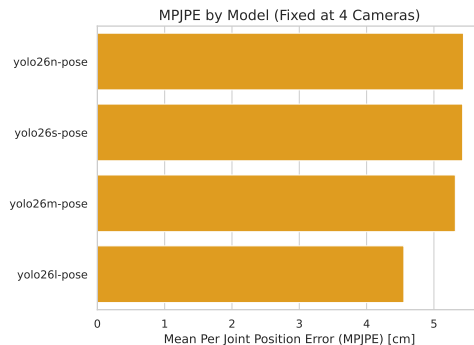
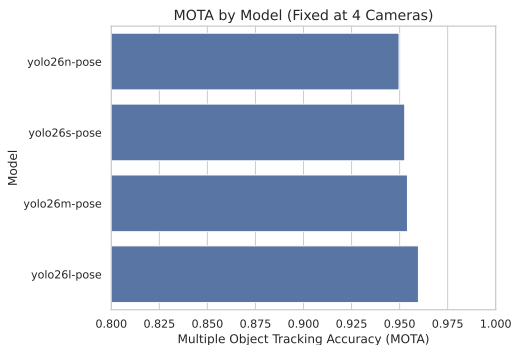
Execution Time Breakdown per Camera Count (Fixed at YOLO26n-pose)



Quantitative Results – Performance vs. Model Size

► Main Finding:

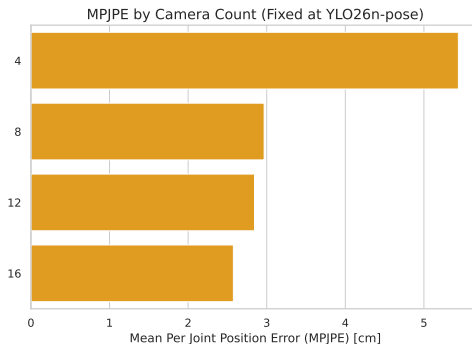
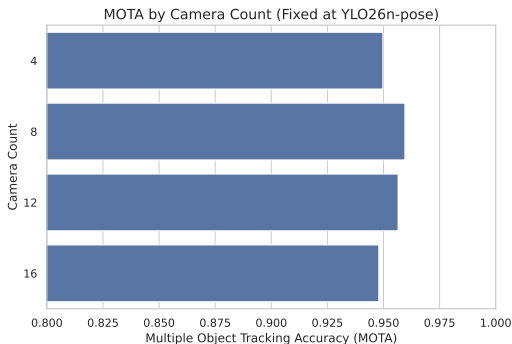
- Larger models improve both MOTA and MPJPE metrics.
- Especially large version gives good results.
- Small and medium version are not significantly better than nano version.



Quantitative Results – Performance vs. Camera Count

► Main Finding:

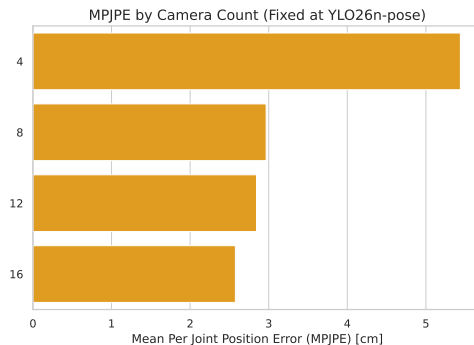
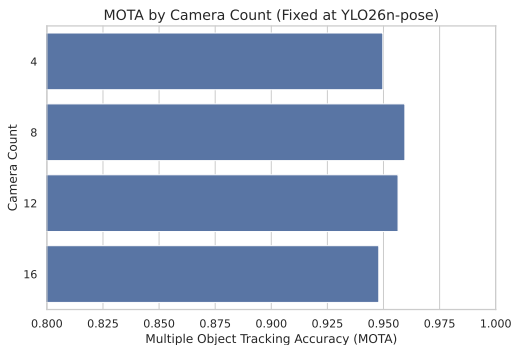
- Adding more cameras significantly improves MPJPE.
- Additional views slightly reduce MOTA due to more false positives.
- More camera views is more effective than larger model capacity for MPJPE.



Quantitative Results – Skeletal Constraints

► Main Finding:

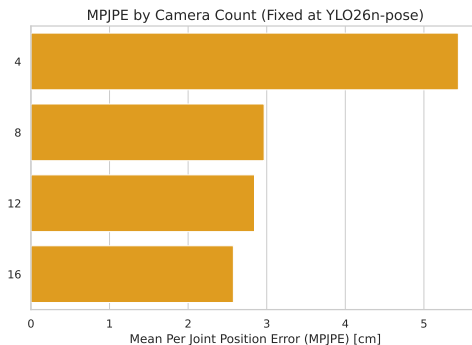
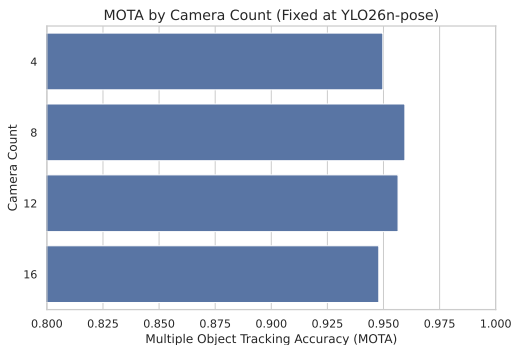
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- Id ut nihil aut et numquam.



Quantitative Results – Learned Parameters

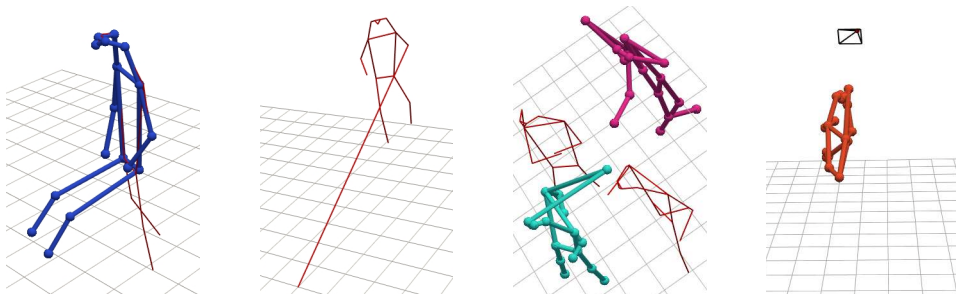
► Main Finding:

- Ipsum laborum incidunt consequatur nobis.
- Consequuntur repudiandae non odio sunt consequatur alias.
- Id ut nihil aut et numquam.



Qualitative Results

- ▶ Common failure modes:
 - Partially occluded persons keypoints drifting.
 - Inactive track seen by only one camera can not cause track creation.
 - Active track seen by only one camera can cause drift.
 - Detection of person while exiting when ground truth is missing.



Conclusion

- ▶ **Summary:**

- Implemented an EKF framework that merges multi-view observations.
- Incorporated skeletal length consistency constraints.
- Implemented simple learning of parameters from data.
- Analyzed performance across different system configuration.

- ▶ **Limitations:**

- Still struggles significantly with occlusions.
- Running 2D pose estimation remains a performance bottleneck.

- ▶ **Possible Future Improvements:**

- Add more constraints that force other known properties, such as joint angles.
- Find more stable learning methods for the Kalman filter parameters.
- Use of a particle filter with occlusion aware likelihood.

